
CPSC 66 Final Report:

Predicting NBA Lineup Efficiency Using Player Archetypes

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Abstract

This project predicts the efficiency of NBA lineups based on combinations of player archetypes. We began by using PCA analysis and K-means++ to identify and define player archetypes within each basketball position. Once we have defined clusters from 2024-25 season statistics, these clusters enable us to label every player according to their attributes by more categories than traditional positions. We then used these labels to predict future lineup efficiency using season data to train and validate a Random Forest model capable of predicting lineup efficiency. Through our analysis we assessed whether player type and combinations of specific attributes can reveal the most efficient NBA lineup. This data-driven analysis reveals the player archetype combinations that will lead to success in the NBA.

1. Introduction

Traditional basketball analysis evaluates players and lineups using the five fixed positions and individual statistics. However, modern NBA success increasingly depends on how different player skill sets fit together, as lineups are more positionless and players contribute in diverse ways that traditional labels fail to capture. This project addresses the question: can player archetypes provide a predictive and interpretable representation of lineup efficiency and success?

To answer this, we designed a two-stage machine learning approach that combines multiple algorithms to model both player characteristics and lineup performance. First, we applied Principal Component Analysis and K-means clustering to identify player archetypes based on each player's individual statistics across the 2024-25 season. These

archetypes capture meaningful playstyle differences among players, such as shooting, defensive impact, and rebounding. We then used these archetype combinations in a Random Forest Model trained on the 2024-25 lineup data to predict the PlusMinus statistic given a set of 5 players. We chose the PlusMinus statistic, which measures points scored minus points against, as the overall measure of efficiency and success of a team.

By combining unsupervised and supervised machine learning techniques within a unified framework, this project illustrates how data-driven approaches can reveal meaningful structure in high-dimensional data.

2. Methods

Our methodology includes three main strategies to make our data interpretable, and generalize trends to make accurate predictions. First, we used Principal Component Analysis (PCA) to reduce the dimensionality of our data. We then used K-Means clustering on our PCA-transformed data to identify player archetypes based on statistical profiles. Finally, after assigning each player to its clustered label, we trained a Random Forest model based on NBA lineups and their PlusMinus scores to generate future predictions of lineup effectiveness.

For reference, we chose to predict the PlusMinus point differential compound statistic as our measure of success for a lineup. In this paper, efficiency refers to the PlusMinus statistic per 100 possessions which is alternatively known as NetRtg (net rating)¹.

2.1. Principal Component Analysis

Because our player-level dataset contains a large number of correlated statistics, dimensionality reduction is a necessary preprocessing step prior to clustering. Reducing the dimensionality of our statistics was important to remove noise, minimize correlated statistics, and increase in-

interpretability. We used Principal Component Analysis to understand how many dimensions were necessary while maintaining as much variance as possible. Given so many statistics from each game, play, and player, we wanted to make sure we were stripping out statistics that were correlated or unnecessary for clustering. Figure 1 depicts the variance from each principal component.

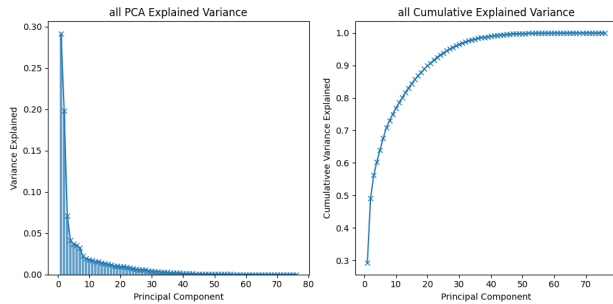


Figure 1. This plot shows the proportion of variance explained by each principal component computed from player statistics across all positions. From this graph it is clear that around 9 principal components explain a majority of the variance in the data.

Figure 1 above shows the PCA for all positions, but it is important to note we ran PCA separately by position before clustering. Because player statistics differ substantially across positions, we perform PCA separately for each position group to ensure that dominant sources of variance reflect role-specific patterns specific to each position. The resulting components are then used for visualization and clustering within each position. Finally, we used PCA to increase interpretability. As shown in Figure 2, we wanted to be able to visualize the clusters and the groups of players that are similar. From reducing dimensionality, we were then able to visualize the clusters more easily in two dimensions.

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2.2. K-Means Clustering

After using PCA to reduce dimensionality, our main method was to use K-Means to cluster the players into archetypes based on their statistics. We used the gap statistic to determine the optimal k on a per-player basis, as this statistic values a stronger clustering structure compared to random data. More specifically, the gap statistic measures how much better the observed clustering compactness is compared to what would be expected under a null reference distribution with no inherent cluster structure. Typi-

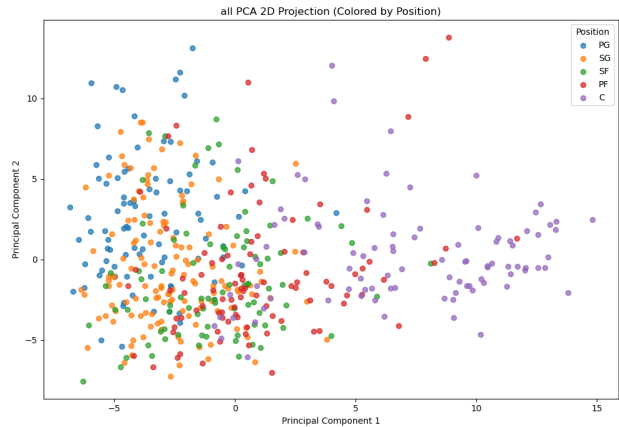


Figure 2. This plot shows all players represented in the first two principal components computed from player statistics across all positions. Each position is color coded

cal methodology using the gap statistic suggests choosing k such that the gap statistic of $k + 1$ clusters is not significantly better than k clusters, as the gap statistic generally rises with k . We found that this k was too small for our clusters to be interpretable, so we averaged this k with the k matching the highest gap statistic for all k less than or equal to ten, our assumption being more than ten archetypes per position is too specific to capture trends with limited lineup data.

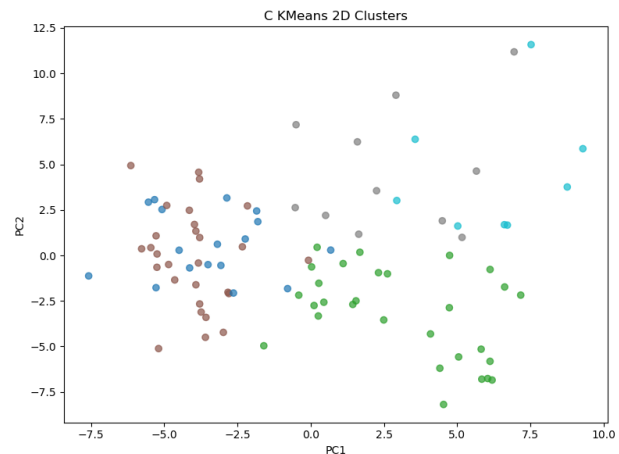


Figure 3. This plot shows the clustering result for the Center position, as represented in two dimensions

To ensure these clusters were interpretable, we printed the top ten players closest to the cluster centers. In addition, checking the statistical features that differ most from one cluster to the others in that position provided insight into why those players were clustered together. This was done

by taking the difference between the standardized statistics for the players in a singular cluster and the mean standardized statistics of the other clusters in a particular position, and seeing which statistical features differed most from the mean in magnitude. Most clusters were defined by one particular skill – such as rebounding or three point shooting – but other clusters outlined players traditionally seen as ‘better’ than others – high points and usage relative to the mean – or ‘worse’ than others – low shooting percentage, defensive rating, etc. For more in depth analysis on the interpretability of the clusters, see the results section.

2.3. Random Forest Model

To evaluate whether player archetypes contain predictive signals for lineup performance, we employed a Random Forest regression model to predict lineup efficiency from archetype composition. Random Forests are ensemble learning methods that combine the predictions of multiple decision trees, each trained on a different bootstrap sample of the data. We believe Random Forest regression is well-suited to this task for several reasons. First, lineup performance may depend on nonlinear interactions between player types that are difficult to capture with linear models. Bootstrapped decision trees naturally model these complex interactions, and the ensemble structure of Random Forests reduces overfitting while maintaining flexibility. Second, Random Forests require minimal feature engineering and are robust to noisy or weakly informative features, which is important given the variability in predicting lineup efficiency.

Each lineup was represented as a fixed-length feature vector representing the archetypes of the players in the lineup. Because NBA lineups often do not have one of each position on the floor at the same time, we adopted a one-vs-all representation in which each feature corresponds to one player archetype, rather than having five features representing the archetype of a specific position. The value of each feature indicates the number of players from that archetype present in the lineup. This approach allows the model to accommodate lineups with unconventional positional configurations, such as multiple guards or positionless lineups, without imposing artificial constraints.

In total, each observation consisted of 25 archetype count features, corresponding to the clusters identified across all positions. Lineup minutes played were not included as a predictive feature but were instead used as sample weights during training and evaluation, ensuring that lineups appearing more frequently in games contributed proportionally to the learning objective. The target variable was the lineup’s efficiency rating.

We trained the Random Forest model using scikit-learn’s RandomForestRegressor. Model performance was evalu-

ated using five-fold cross-validation, with each fold preserving the full distribution of lineup observations. Within each training fold, hyperparameters were tuned using grid search to identify configurations that maximized predictive performance. The hyperparameters considered included the maximum depth of each tree, the number of features considered at each split, the minimum number of samples required at leaf nodes, and the total number of trees in the ensemble.

For each cross-validation fold:

- The model was trained on 80% of the data, with lineup minutes used as sample weights.
- Tune hyperparameters via grid search
- Evaluate weighted R^2 on held-out fold
- Aggregate performance across folds

2.4. Analysis of Methods

We employed clustering to identify patterns in player types from statistics to move beyond traditional position categorization. By grouping players with similar statistical profiles into archetypes, PCA and K-Means clustering provides an interpretable representation of player roles such that we can identify the most effective skill sets for on-court performance. In particular, clustering exposes statistical dimensions that most differentiate player roles. This abstraction aligns with important considerations of coaches when constructing lineups. It is important to consider complementary roles and skills when laying out a team.

We chose the Random Forest model because it can capture complex, nonlinear interactions between player archetypes that influence lineup performance. The ensemble nature of Random Forests reduces overfitting while maintaining flexibility to model complex data.

3. Experiments and Results

3.1. Data Collection and Preprocessing

We used two distinct NBA datasets aggregated from official NBA box scores and play-by-play data. The first dataset, **NBA Player Stats: Per 100 Possessions (2024-25, 2025-26)**, contains season-level individual player statistics and is used to identify player archetypes. The second dataset, **PBP NBA Lineup Stats (2024-25, 2025-26)**, contains season-level statistics for five-man lineups and is used to model lineup efficiency. What we are referring to as overall team efficiency is also known as “PlusMinus.” PlusMinus is a composite statistic that subtracts points against from points for during the time that any given lineup is on

the court. This values the strength of a lineup, and to predict a lineup’s PlusMinus output will give us valuable insight into the future success of a team.

All analysis was conducted using Python, with sci-kit learn libraries for preprocessing, clustering, and modeling. Prior to analysis, we applied several preprocessing steps to ensure the data was consistent and interpretable. First, player and lineup features were measured on different scales, with some recorded as raw counts and others as percentages. To prevent scale-driven bias, all numerical features were standardized. Second, the individual player dataset included full names, while the lineup dataset listed only last names, preventing direct alignment between the two. We addressed this by writing a Python script to match last names to full player names, enabling consistent player identification and allowing archetype labels to be correctly assigned to each five-man lineup. Finally, we removed overly specific and redundant features, such as highly granular shot-type statistics, to reduce noise, improve computational efficiency, and increase model stability. These steps helped ensure our analysis reflected real player roles and lineup behavior, not distortions caused by data formatting or scale

3.2. Clustering Results

To interpret the resulting player archetypes, we examined each cluster by identifying (1) the number of clusters and the amount of players per cluster (2) the players closest to the cluster centroid and (3) the statistics that most strongly differentiated each cluster from the others. Positive values indicate features that are more associated within a cluster, while negative values indicate features that are less associated relative to other clusters.

As an example, we consider the point guard (PG) position, for which four distinct clusters were identified, with the following sizes:

- Cluster 0: 34 players
- Cluster 1: 28 players
- Cluster 2: 22 players
- Cluster 3: 16 players

Table 1 summarizes the characteristics of Point Guard Cluster 0. Table 2 summarizes the characteristics of Point Guard Cluster 1. We will interpret these two clusters’ characteristics as an example of the value of our clustering approach.

These features suggest that Cluster 0 is characterized by point guards who rely heavily on three-point shooting, as evidenced by elevated three-point attempt percentage and assisted three-point scoring. Conversely, players in this

Table 1. Summary of Point Guard Cluster 0 archetype characteristics.

Attribute	Cluster 0 (PG)
Number of Players	34
Representative Players	Reggie Jackson, Mike Conley, Jose Alvarado, Aaron Holiday, Marcus Smart, Cameron Payne, Marcus Sasser, Keaton Wallace, Monté Morris, Fred VanVleet
Dominant Positive Features	FG3APct (+1.96) AtRimFG3AFrequency (+1.82) PtsAssisted3s (+1.73)
Dominant Negative Features	ShortMidRangeFrequency (−1.97) AtRimFGA (−1.87) PtsAssisted2s (−1.86) DefThreePtReboundPct (−1.68) PtsPutbacks (−1.66)
Archetype Interpretation	Perimeter-oriented point guards emphasizing three-point shooting and assisted perimeter scoring, with limited rim pressure and rebounding contribution

cluster tend to take fewer shots at the rim, generate fewer assisted two-point baskets, and contribute less to rebounding and putback scoring. This interpretation is reinforced by the players nearest to the cluster centroid, who are predominantly smaller, perimeter-oriented guards. While many of these players are capable shooters, they generally do not score across a wide variety of locations, instead specializing in perimeter-based roles.

Point Guards in cluster 1 contribute significantly to rebounding, shooting poorly from behind the three point line and taking less shots from long distances. Instead, these guards shoot high quality shots from short range and grab offensive rebounds. This is supported by the players closest to the cluster center – mostly taller and guards who use their strength, speed, or vertical jump to get close to the basket. It’s easy to see that though these players play the same position as those in PG Cluster 0, these are two vastly different styles of play.

Not just for Point Guards, but across all 25 clusters spanning the five positions, we found that each cluster was highly interpretable and exhibited clear stylistic distinctions. Several clusters aligned closely with well-established basketball archetypes. For example, Center Cluster 3 consisted primarily of tall, athletic centers who specialized in rim protection and finishing efficiently near the basket. Shooting Guard Cluster 0 was defined by players who emphasized three-point shooting and perimeter defense, reflecting a classic “3-and-D” role. Small Forward Cluster 2 captured larger wings who prioritized attacking the rim and generating offense through drives rather

Table 2. Summary of Point Guard Cluster 1 archetype characteristics.

Attribute	Cluster 1 (PG)
Number of Players	23
Representative Players	Anthony Black, Killian Hayes, Devin Carter, Terry Rozier, Jrue Holiday, Bruce Brown, KJ Simpson, Craig Porter Jr., Scotty Pippen Jr., Kris Dunn
Dominant Positive Features	OffRebounds (+1.99) OffFGR reboundPct (+1.99) OffThreePtReboundPct (+1.99) OffThreePtRebounds (+1.98) ShotQualityAvg (+1.93)
Dominant Negative Features	LongMidRangeAccuracy (-1.96) NonHeaveFg3Pct (-1.93) Avg2ptShotDistance (-1.93) UnblockedCorner3Accuracy(-1.90) Corner3Accuracy (-1.89)
Archetype Interpretation	Rebounding-focused players who generate value through offensive boards and high-quality shots near the basket. These players are bigger, more physical guards who are known for their athleticism.

than perimeter shooting. Finally, Power Forward Cluster 4 was composed of smaller, floor-spacing forwards who frequently operated from the corners and stretched opposing defenses with their three-point shooting.

Notably, for every position, at least one cluster consisted primarily of elite, high-usage players. (EX PF Cluster 0: Paolo Banchero, Scottie Barnes, Paul George, Kevin Durant, Jayson Tatum) Clusters of this type were consistently differentiated from other clusters by substantially higher scoring-related statistics. The emergence of such clusters provides additional validation that the clustering approach successfully captured meaningful differences in player roles and performance profiles.

To support this assertion, we calculated the average PlusMinus for each cluster to gain insight into archetypes that have significantly positive or negative effects on lineup performance. Unsurprisingly, clusters that included elite, high-usage players were in the positives, but some archetypes that had flown under radar were shown to contribute significantly to efficient lineups. For example, Centers that spaced the floor by shooting from the perimeter, and particularly the corner (Center cluster 1), had the highest average PlusMinus across lineups with 7.42. One possible reason for this may be that spot-up shooting Centers open up space for other players to cut and drive to the basket, raising the effectiveness of other players in their line-

ups. This affirmed our belief that there are underlying relationships between archetypes that contribute to lineup efficiency, relationships we hoped to be capturable through certain regression models.

To further see effects on lineup efficiency by individual clusters, we trained a linear regression model and examined the learned coefficients. While the linear regression model performed worse than the Random Forest, the coefficients provide a direct interpretation of archetype contributions to lineup efficiency, representing the marginal change in lineup PlusMinus (per 100 possessions) associated with adding one additional player from a given positional cluster and holding the rest of the lineup composition fixed. Interestingly, Center cluster 1 had the fifth highest coefficient of 2.99, with the highest coefficient (4.99) belonging to Center cluster 4 – Centers who scored a lot from the perimeter and free throw line but also achieved a high number of assists. One unexpected result was Shooting Guard cluster 3 – a cluster of guards different from other archetypes by their rebounding on both ends of the floor – which achieved the second highest weighted average PlusMinus (6.99) and second highest learned coefficient (4.94). This is surprising, as rebounding isn’t typically considered a high priority for Shooting Guards, suggesting this trait may be undervalued for the position.

3.3. Random Forest Results

Across cross-validation folds on the 2024–25 data, the archetype-based Random Forest achieved a mean weighted R^2 of 0.11 (standard deviation 0.06). The R^2 metric shows the proportion of the variance in our PlusMinus label that can be explained by the cluster features. A low R^2 indicates that the model does not fit the data well, and does not perform significantly better than predicting the mean PlusMinus. While predictive performance varied across folds, an R^2 of 0.11 demonstrates that the model consistently explained a non-trivial portion of variance in lineup efficiency relative to a constant baseline. The Random Forest also achieved a mean weighted mean squared error (MSE) of 213.6 (standard deviation 23.3). This corresponds to an average root mean squared error of approximately 14.6 points per 100 possessions, meaning our efficiency predictions were off by roughly this estimate.

When the same model was applied to 2025–26 lineups, performance deteriorated substantially. Cross-season evaluation yielded a negative weighted R^2 (0.02) and a mean weighted MSE of 725.9 (standard deviation 183.5), corresponding to a root mean squared error of roughly 27.0 points per 100 possessions. Despite attempts to constrain model complexity by limiting tree depth and increasing the minimum number of samples per leaf, we were unable to increase the 2025–26 R^2 beyond 0.02. These re-

sults indicate limited generalization across seasons when using archetype-based Random Forest predictions, suggesting that lineup efficiency is influenced by substantial season-specific factors not captured by archetype composition alone.

To investigate whether information loss from dimensionality reduction and clustering contributed to the limited predictive performance, we conducted a parallel experiment using raw player statistics rather than archetypes. In this setting, each player in a lineup was represented using a reduced set of 15 per-100-possession statistics selected to balance expressiveness and redundancy. Lineup feature vectors were constructed by concatenating individual player statistics, and the same Random Forest training and evaluation procedure was applied.

The raw-statistics model achieved a mean weighted R^2 of 0.08 during cross-validation on 2024–25 data, comparable to the archetype-based model. When applied to 2025–26 lineups, the raw-statistics model achieved a barely positive weighted R^2 (0.03) and higher weighted correlation (0.24) relative to the archetype-based approach, though overall predictive performance remained limited.

Together, these results suggest that neither archetype-based and raw-statistic Random Forest approaches generalizes strongly across seasons. Importantly, the comparable performance of the two models indicates that clustering and dimensionality reduction did not result in substantial information loss. Instead, the limited predictive accuracy likely reflects the complexity of predicting lineup efficiency, which is influenced in significantly more ways than the player's play styles or stats.

4. Social Implications and Discussion

The key stakeholders in this project include NBA coaches, front-office decision makers, players, and fans and media analysts. Coaches and analytics staff stand to benefit directly from clearer insights into player archetypes and lineup efficiency, which can support decisions around rotations, roster construction, and mid-game adjustments. Front offices may use these results to inform drafting, trades, and contract valuation by identifying undervalued and overvalued skill combinations, as well as complementary player profiles. Players could benefit if the analysis highlights the value of subjective and off-ball impact, but they could also be disadvantaged if metrics or outputs oversimplify their role. Fans and media analysts may gain a deeper understanding of why certain lineups succeed and may see their favorite team succeed because of these analyses, though there is also a risk of overemphasizing styles of play approved by the model at the expense of creativity or team identity.

This work interacts with existing power dynamics by potentially shifting influence further toward data and analytics within basketball organizations, which may empower analytically inclined staff while marginalizing experiential knowledge from coaches or scouts if used uncritically. There are also real concerns about misuse: models trained on historical data may reinforce existing biases (e.g., undervaluing certain positions, play styles, or players with smaller sample sizes) and if model outputs are treated as definitive causal claims rather than decision-support signals with uncertainty, they can lead to overconfidence, misallocation of playing time or salary, and unfair evaluations of players.

From an economic and ethical perspective, these models could influence salaries, playing time, and career longevity, raising questions about transparency, accountability, and fairness. Rather than serving as a definitive decision-maker, this project is best viewed as a decision-support tool whose value depends on careful integration with domain expertise, awareness of limitations, and ongoing evaluation of how its outputs affect both competitive balance and individual stakeholders.

5. Conclusions

Despite limited predictive performance, one of the most compelling outcomes of this work was the interpretability of the learned archetypes. The clustering approach consistently produced intuitive groupings in ways that could be useful for similar projects looking to evaluate player archetypes.

This project does yield questions and future work. First, incorporating contextual information such as opponent lineups, game state, or possession-level data could improve predictive performance by capturing factors that strongly influence lineup outcomes. Second, more complex models like neural networks or more advanced architectures beyond the scope of this class may better capture some of the nonlinear relationships between players. It is important to note though, that such approaches may come at the cost of interpretability. Overall, this project demonstrates both the strengths and limitations of archetype-based modeling for lineup analysis.

Acknowledgments

We would like to thank Professor Gabriel Hope for his support throughout the work for this paper. We would also like to acknowledge the help provided by student feedback from our peers. Thank you to those who provided helpful comments during the last stages of our work. Our work is supported by the Swarthmore College Computer Science department.

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